Data Visualization Using Power BI

Aim

To learn the Power BI interface, connect to various data sources such as Excel, CSV, and SQL databases, create basic visualizations using bar charts, line charts, and pie charts, develop calculated columns and measures, and build an interactive dashboard for data analysis.

Procedure

1. Open Microsoft Power BI Desktop and explore the interface, including the Report, Data, and Model views.
2. Import data from Excel, CSV, or SQL databases using Home → Get Data.
3. Select the required file or database, preview the data, and click Load. Use Transform Data to clean or modify data if necessary.
4. Select the required fields from the Data pane to create visualizations.
5. Create a bar chart to compare values across categories.
6. Create a line chart to display trends over time.
7. Create a pie chart to represent proportions of different categories.
8. Create a calculated column using Modeling → New Column.
9. Create a measure using Modeling → New Measure to calculate aggregated values.

10. Format the visualizations by adding titles, labels, legends, and appropriate colors.
11. Arrange the charts and other visuals on a report page to create a dashboard-style view.
12. Publish the report to Power BI Service, if available, and pin visuals to a dashboard when dashboard functionality is required.

Sample Dataset

Create an Excel file named SalesData.xlsx with the following data, or enter the data into Excel and save it as a CSV file.

Date	Product	Region	Sales	Quantity
01-01-2026	Laptop	Chennai	50000	2
05-01-2026	Mobile	Madurai	30000	3
10-02-2026	Laptop	Chennai	60000	2
15-02-2026	Tablet	Coimbatore	20000	2
20-03-2026	Mobile	Madurai	40000	4
25-03-2026	Tablet	Chennai	25000	1

Connecting to Data Sources

A. Excel file

- Select Home → Get Data → Excel Workbook.

- Choose SalesData.xlsx.
- Select the required worksheet and click Load.

B. CSV file

- Select Home → Get Data → Text/CSV.
- Choose the CSV file.
- Check the preview and click Load.

C. SQL database

- Select Home → Get Data → SQL Server.
- Enter the server name and database name.
- Select the required table and click Load.
- Enter database credentials if prompted.

Creating Visualizations

A. Bar chart

1. Select the Clustered bar chart visual.
2. Drag Product into the Y-axis category field.
3. Drag Sales into the X-axis value field.
4. Add the title Sales by Product.

B. Line chart

1. Select the Line chart visual.
2. Drag Date into the X-axis field.
3. Drag Sales into the Y-axis field.

4. Set the date axis to the desired date hierarchy or continuous format.
5. Add the title Sales Trend Over Time.

C. Pie chart

1. Select the Pie chart visual.
2. Drag Product into the Legend field.
3. Drag Sales into the Values field.
4. Enable data labels to display the proportions.
5. Add the title Sales Distribution by Product.

Creating Calculated Columns and Measures

A. Calculated column

Select Modeling → New Column and enter:

Total Amount = SalesData[Sales] * SalesData[Quantity]

This creates a new column by multiplying sales by quantity for each row.

B. Measure: Total Sales

Select Modeling → New Measure and enter:

Total Sales = SUM(SalesData[Sales])

C. Measure: Total Quantity

Total Quantity = SUM(SalesData[Quantity])

D. Measure: Average Sales

Average Sales = AVERAGE(SalesData[Sales])

Building a Dashboard

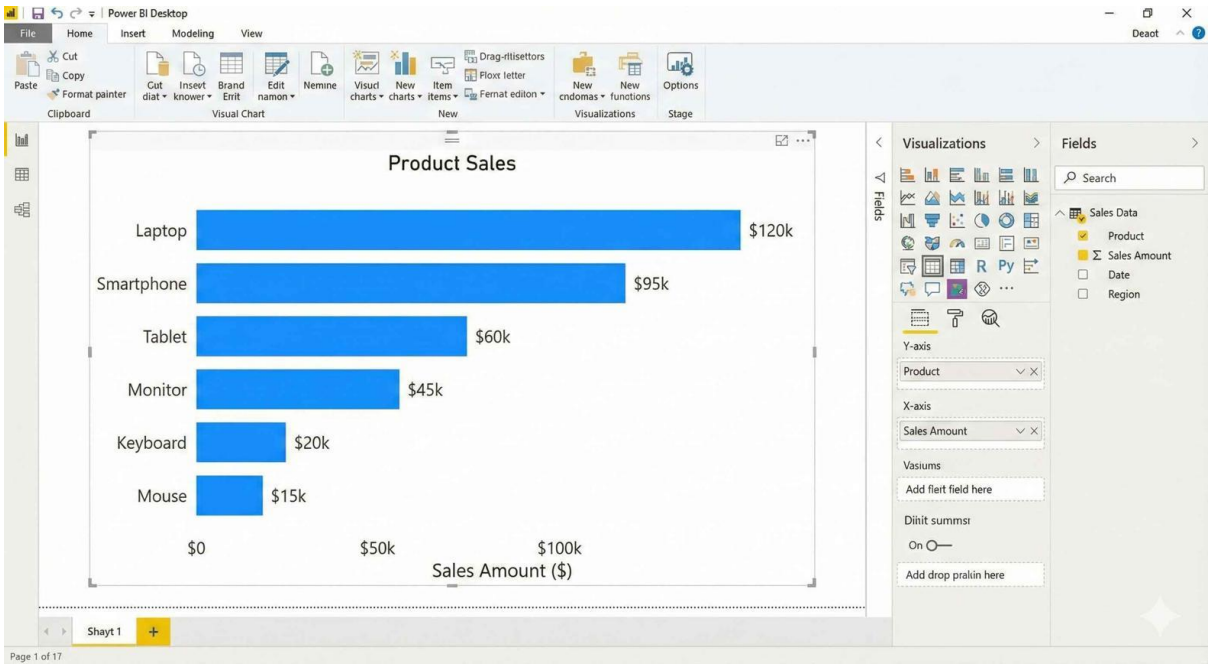
Sample Power BI Sales Dashboard

Total Sales

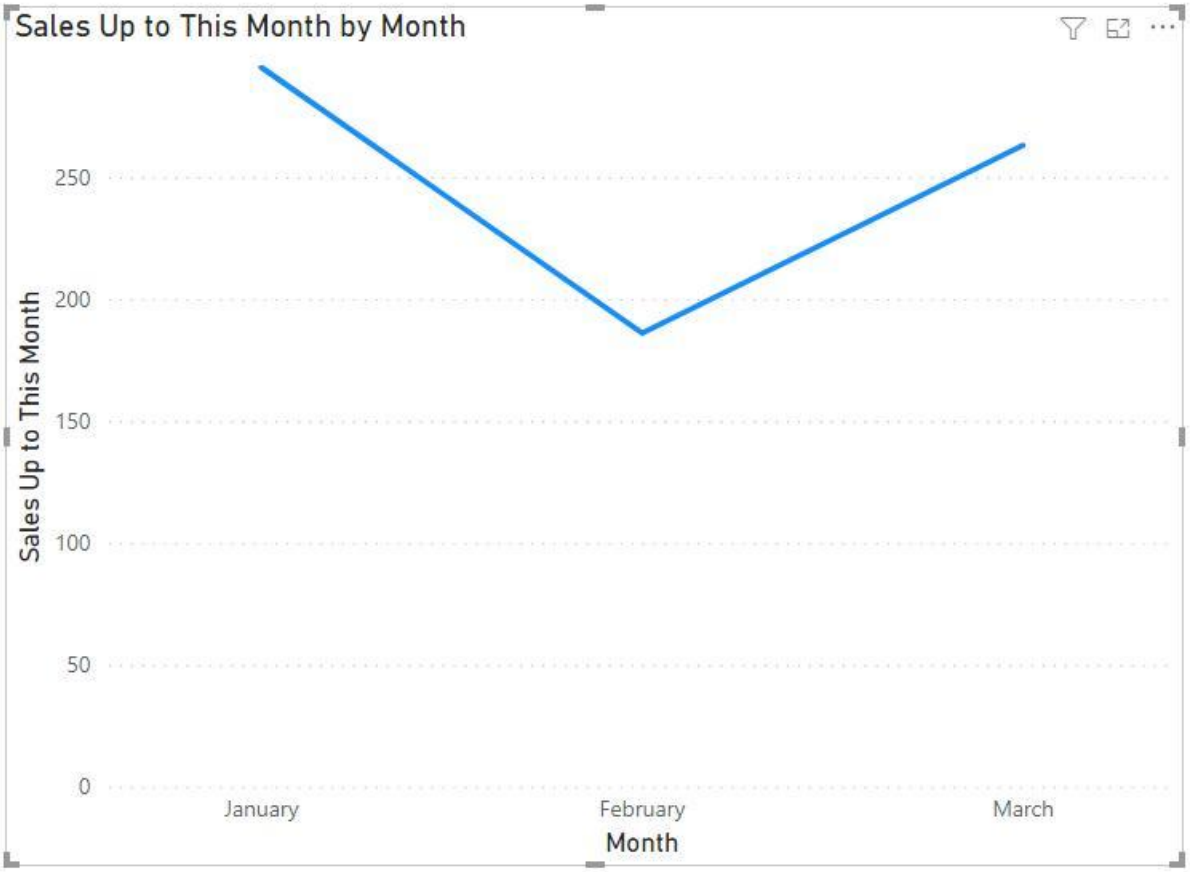
₹2,25,000

Total Quantity

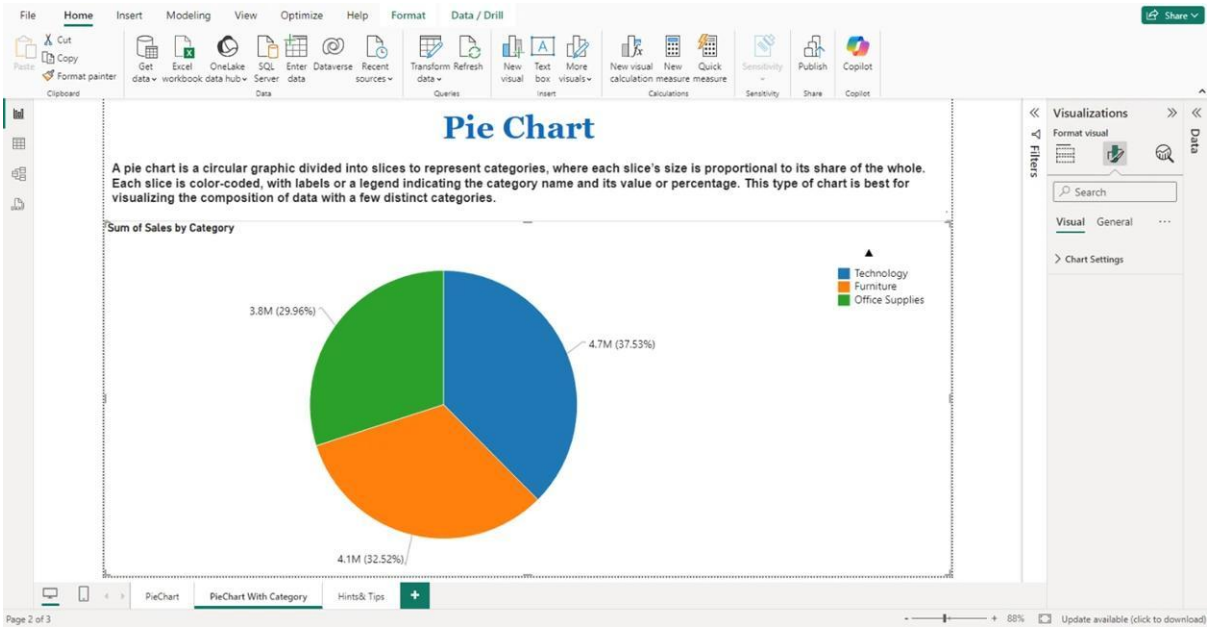
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Sales by Product



Sales Trend



Sales Distribution by Product

Steps to arrange the report:

1. Add Card visuals for Total Sales and Total Quantity.
2. Place the bar chart, line chart, and pie chart on the report canvas.
3. Add a slicer using Region to filter the report by location.
4. Format titles, labels, and data values for readability.
5. Save the report as SalesDashboard.pbix.
6. To create a true Power BI Service dashboard, publish the report and pin the required visuals to a dashboard if your account and workspace support this feature.

8. Output

The expected output is a Power BI report containing:

- A bar chart comparing sales across products.
- A line chart showing the sales trend over time.
- A pie chart showing each product's share of sales.
- Calculated columns and measures displaying derived values and summary statistics.
- An organized dashboard-style report with interactive filtering and summary cards.

Result

Thus, data visualization using Microsoft Power BI was studied successfully. Data was imported from Excel, CSV, and SQL database sources. Bar charts, line charts, and pie charts were

created to represent and analyze the data. Calculated columns and measures were developed using DAX, and the visualizations were arranged into a dashboard-style report for effective data analysis and decision-making.